# Securing understanding of arithmetic sequences

**1** In the $n^{th}$ term expression $9n - 7$, you call $9$ the ___________ of $n$. Tick **1** correct answer

- [ ] $1^{st}$ term
- [x] coefficient
- [ ] equation
- [ ] expression

**2** What is the common difference in the arithmetic sequence $6n - 3$? Tick **1** correct answer

- [ ] -6
- [ ] -3
- [ ] 3
- [x] 6

**3** Match each sequence to its $n^{th}$ term. Write the correct letter in each box

| | |
|---|---|
| a | $3n + 9$ |
| b | $9n + 3$ |
| c | $9n - 3$ |
| d | $3n - 9$ |
| e | $9 - 3n$ |

| | |
|---|---|
| a | $12, 15, 18, 21, 24, \ldots$ |
| e | $6, 3, 0, -3, -6, \ldots$ |
| d | $-6, -3, 0, 3, 6, \ldots$ |
| c | $6, 15, 24, 33, 42, \ldots$ |
| b | $12, 21, 30, 39, 48, \ldots$ |

**4** 405 is a term in the arithmetic sequence with the rule $7n - 50$. What is its position in the sequence? $n =$ 65 Fill in the blank

**5** Is $189$ a term in the arithmetic sequence $7n - 50$? Tick **1** correct answer

- ☐ Yes, because $189$ is odd and the sequence contains a lot of odd numbers.
- ☐ Yes, because $189 \div 7 = 27$
- ☐ No, because $189$ is odd and the sequence is mostly even.
- ☑ No, because $7n - 50 = 189$ has a non-integer solution.

**6** Which expressions could be used to generalise the arithmetic sequence 5, 9, 13, 17, 21, …? Tick **2** correct answers

- ☐ $5n + 4$
- ☑ $4n + 1$
- ☐ $5 + 4n$
- ☑ $5 + 4(n - 1)$
- ☐ $4 + 5(n - 1)$